Oct 19 — What we did today

IAA AI-Assisted Hackathon 2026 · Woojin Ji & Krista Stanke · Part 1 closes Friday, Oct 23

> **In one line.** We made our first Kaggle submissions and moved the public score from **0.11803 to 0.11973** (adopted model v7). Along the way we corrected the target definition, found a strong signal in past claims, and made every decision with **rules fixed before seeing the results**. At the end we found the likely cause of motor under-pricing (the bodily injury severity trend).

0. The big picture

The competition asks for the **expected ultimate claim cost of each 2025 policy**, net of excess and recoveries, excluding declined claims. It is the **risk premium**, with no expenses or profit.

Our model has two steps.

- **Step 1, build the target:** bring 2021–2024 claims to 2025 level (severity trends, the 2023 bodily injury reform, frequency shifts). Recent, less-reported years get less weight; very large claims are capped and handled separately.
- **Step 2, a model per line:** predict the rate per unit of exposure from policy features (driver age, region, weather, past claims, ...) and multiply by 2025 exposure.

> **How to read the score (D^2).** A single policy's loss is mostly chance (usually zero, sometimes large), so even a perfect model scores around 0.1. The top 10 teams were at 0.125–0.130. Differences of 0.001 can be noise.

1. Score journey

Version	Change	Public score	In plain words
v2	Regularised GLM, balance correction	0.11803	First submission, but with the wrong excess definition
B	Excess deducted on every peril	0.11724	Correct definition, lower score (motor level went down)
v3	Unstable features removed	0.11798	Removing noise recovered the score
v4	Claim history added	0.11907	The clearest gain of the day
v5	Contents GLM + GBM blend	0.11996	First GBM to pass our rule
v6	Large-loss loading from a fitted tail; feature clean-up	0.11951	Sounder, slightly lower score (first prediction miss)
v7 (adopted)	**Contents: better-regularised GBM, 3-seed average**	**0.11973**	**Current final candidate**
blend v1 / v2	Experiments, not adopted	0.12113 / 0.12186	Higher score, but over-price the safest customers

2. What we did, step by step

Each item: Why Found Decided

① Regularisation and balance correction

Why The first model under-priced the safest policies by about 20%: relativities were too steep.

Found Ridge, which pulls relativities toward the average, fixed it. But pulling them in also lowers the total (2× and 0.5× pulled to 1.4× and 0.7× turn a total of 250 into 212). One factor k, set on training data, restores the level. **Like archery**: ridge tightens the grouping; k adjusts the sight.

Decided Motor α 0.5 / 10 bands, buildings α 0.5 / 20, contents α 0.3 / 20. Validation D^2: 0.050→0.051, 0.055→0.065, 0.060→0.067.

② Excess definition (confirmed by the host)

Why The data could not tell whether the excess applies to liability perils. AI had assumed it does not (UK practice).

Found The host answered on the forum: it applies to every peril. **Yet the leaderboard scored the wrong definition 0.0008 higher.** Trusting the leaderboard would have locked in the wrong target.

Decided Excess on every peril. Default changed in `target.py`; five test expectations recalculated by hand.

③ Feature checks: stability and links

Why The GBM ranked "previous insurer" as a top feature, which made little actuarial sense (we also tested an adverse-selection explanation).

Found The ranking of previous insurers by cost did not hold between 2021–22 and 2023–24 (rank correlation ≈ 0): the model had memorised chance. We also found renewal count = customer tenure (identical), age ≈ years licensed, region ≈ weather, occupation ≈ age.

Decided Removing features without stable signal, and the duplicate, improved the GLM (v3).

④ Claim history (the biggest finding)

Why Past claims are the best predictor of future claims in insurance, and we were not using them yet.

Found Policies with a claim in the previous year were **under-priced by 22–80%**. The customer-stated prior claims field was almost unrelated to recorded claims (correlation 0.04).

Decided History built from claims reported before 1 January of the policy year, the same information a 2025 test row has; unknown history kept as "unknown", not zero (v4, +0.0011). History by peril and past claim **amounts** were not used (no gain; amounts would leak later information).

⑤ Testing a GBM (LightGBM)

Why The gap to the top (about 0.01) may come from interactions a GLM cannot capture.

Found +0.007 to +0.011 validation D^2 in buildings and contents, but the safest 10% of policies were **priced over 30% too high**. 18 settings and a calibration step did not fix buildings. Motor gained almost nothing.

Decided GBM used only in **contents**, which passed the rule (+0.003 or more, tails no more than 0.1 worse) (v5 → v7). Buildings is a team decision on Thursday.

⑥ Large-loss loading: one claim's problem

Why It started from the question "why not just drop very large claims?"

Found **96% of the motor loading (16)** came from one open bodily injury claim of 11.9m; without it, 0.71. A Pareto tail fitted to 119 claims above 100k gave α ≈ 2 and a loading of 2.83, stable across thresholds.

Decided Loading 2.83, allocated in proportion to risk (v6); uncertainty range 1.2–3.8 recorded. Not dropped, because both the real cost and the score include large claims.

⑦ Basic checks: limits and cover

Found Claims above the sum insured: 0.02–0.03%. Accidental damage claims without the add-on were paid claims, so they belong in the target.

Decided No action; recorded.

⑧ Bodily injury severity trend (last finding; first job tomorrow)

Why Every time we lowered motor, the leaderboard score fell: a sign that motor is priced too low.

Found On a like-for-like comparison, BI severity still rises **+7.7% a year (median +9.4%)** after the reform; we had assumed 0%. Changing the trend alone made 2024 validation worse, because the reform factor 0.83 had been estimated assuming zero trend: the two are tangled.

Decided Kept for now (rule not met). Tomorrow: change trend (6%) and reform factor (about 0.79) **together** and validate.

3. How we decide

- **Rules first, results second.** Example: "a GBM is adopted only if D^2 is at least 0.003 above the GLM and the tails are no more than 0.1 worse." Changing the bar after seeing results would fit the bar to the results.
- **Decide on validation; use the leaderboard to check.** We validate on years not used for training (2023, 2024). Before each submission we wrote down the expected change: **6 of 7** were right, and the one miss (v6) could be explained.
- **No fitting to the leaderboard.** Public scores use only part of the test data; choosing by public score can drop a team in the final (private) ranking.

4. How we checked AI (today's examples)

AI output	Outcome
Assumed no excess on liability perils	wrong host answer
Re-checked frequency counting only claims with net > 0	false alarm three definitions compared; original 0.93 confirmed
Tie-break rule direction; a 3% limit on k with no evidence; level check against a four-year average	corrected
Suggested past claim amounts as a feature	withdrawn amounts leak later information
Run-time estimates; v6 score prediction	wrong recorded
Balance correction, claim history, tail loading, leaderboard predictions	confirmed by validation and the leaderboard

Human questions also changed the direction: "Isn't rebalancing artificial?", "Could it be adverse selection?", "What does occupation actually affect?", "Why not drop extreme claims?", "Shouldn't we have checked correlations before modelling?"

5. Current model (v7)

Line	Model	Notes
Motor	Regularised Tweedie GLM (10 bands)	Claim history; home-type features removed; large-loss loading 2.83 allocated by risk
Buildings	Regularised Tweedie GLM (20 bands)	Claim history
Contents	LightGBM (7 leaves, min 5,000, L2 50; 3-seed average)	Claim history; +0.0106 validation D^2 over the GLM

6. The three biggest open problems

1. **Motor is priced too low.** Motor is 56% of the cost, so it moves the score most. First attempt: fix the BI trend and reform factor together.
2. **The buildings GBM gain (+0.009) is unused.** A choice between score and fair prices for the safest customers: team decision on Thursday.
3. **No reproducible pipeline yet.** v7 was produced by running notebook cells in order. This is part of the 20% trust criterion and essential for Week 2.

7. Plan to Friday

Day	Work
Tue (5 h)	BI trend + reform factor; **report outline → share with Krista**; contents GBM p = 1.4; glass trend; contents 2023
Wed (3 h)	pipeline.py, pre-submission checks, automated tests on GitHub
Thu (5 h)	Fix the final model; rebuild with the pipeline and compare; GOVERNANCE.md; diagnostics notebook; **7 pm Zoom with Krista**
Fri (Oct 23)	Final submission; select the two final entries by validation

8. Glossary

Term	Meaning
A/E	Actual ÷ expected. Above 1 = under-priced; below 1 = over-priced
D^2 (Tweedie, p = 1.5)	How much the model reduces error against a single average rate. The competition score
Lowest / highest decile	The 10% of policies with the lowest / highest predicted risk
GLM / GBM	A model that multiplies relativities per feature / a sum of hundreds of decision trees
Ridge	Pulls relativities toward the average so that chance has less effect
Balance factor k	Makes predicted totals on training data equal actual totals
Loading	The expected cost of large claims above the cap, added back
Completeness	Share of a year's claims reported so far; lower for recent years